

Bounds for triple separation in the Witt design

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Abstract

Let $m(\mathcal{W})$ be the least number of octads of the Witt design $S(5, 8, 24)$ whose containment signatures distinguish its 2,024 point triples. We prove $90 \leq m(\mathcal{W}) \leq 127$. The lower bound combines a charge from signatures of weight two with second moments of point and point-pair degrees. The upper bound is given by an explicit family of 127 octads, with a short exact verifier for the design axioms and all triple signatures. The zero signature is allowed throughout. This is a partial result: the exact minimum is not determined, and no priority claim is made.

1 The invariant and the result

Write $V = \{1, \dots, 24\}$. A Steiner system $S(5, 8, 24)$ is a family of eight-element subsets of V in which each five-element subset lies in exactly one block. Its blocks are called *octads*. Let $\mathcal{W}$ be the Witt design; we use its stipulated uniqueness up to point relabeling when transferring the explicit construction below. For a family $\mathcal{F} = \{B_1, \dots, B_m\} \subseteq \mathcal{W}$, define

$$\sigma_{\mathcal{F}}(T) = (\mathbf{1}_{T \subseteq B_i})_{i=1}^m, \quad T \in \binom{V}{3}.$$

The family *separates triples* if these signatures are pairwise distinct. We seek the minimum possible m , denoted by $m(\mathcal{W})$. The zero signature is permitted; no covering or symmetry condition is imposed on $\mathcal{F}$.

Theorem 1. *The triple-separation number of the Witt design satisfies*

$$90 \leq m(\mathcal{W}) \leq 127.$$

The lower bound holds for every $S(5, 8, 24)$. The upper bound is realized by the labeled family specified in Appendix A.

There are $N = \binom{24}{3} = 2024$ triples, so counting binary signatures alone gives $m \geq \lceil \log_2 N \rceil = 11$. The fixed row weight 56 of the octad–triple containment matrix gives a stronger elementary bound. If a_d denotes the number of triples in exactly d selected octads, injectivity gives $a_0 \leq 1$ and $a_1 \leq m$, whence

$$56m = \sum_d da_d \geq a_1 + 2(N - a_0 - a_1) \geq 2N - 2 - m.$$

Thus $m \geq 71$. The structural lower bound in Theorem 1 uses the four-point intersections of octads to restrict how many signatures can have weight two. Together with the explicit upper certificate, it gives an interval for the original invariant without assuming transitivity or selecting particular orbits.

Context and literature scope. Triple degrees in uniform set families provide adjacent context; see Zeng [1]. Exact binary-matrix constructions for finite extremal bounds, such as [2], provide adjacent computational methodology. Neither reference is used as a theorem about $m(\mathcal{W})$. The title *Eight octads suffice* was supplied as a literature lead concerning defining sets. A defining-set question asks whether selected blocks determine a design, whereas the invariant here distinguishes triples inside a fixed design. A verified primary-source comparison for this invariant is unavailable here; neither absence of earlier equivalent bounds nor the current global status of the exact-minimum question is asserted.

2 A universal lower bound

Fix a separating family $\mathcal{F} \subseteq \mathcal{W}$ of size m . For a triple T , let $d(T) = |\{B \in \mathcal{F} : T \subseteq B\}|$ and $a_d = |\{T \in \binom{V}{3} : d(T) = d\}|$.

2.1 Design counts and charges

Counting pairs consisting of a five-subset and its containing block shows that each fixed j -subset lies in

$$\lambda_j = \frac{\binom{24-j}{5-j}}{\binom{8-j}{5-j}} \quad (0 \leq j \leq 5)$$

octads. In particular,

$$(\lambda_0, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = (759, 253, 77, 21, 5).$$

Consequently $0 \leq d(T) \leq 21$, and

$$\sum_{d=0}^{21} a_d = N, \quad \sum_{d=0}^{21} da_d = 56m, \quad a_0 \leq 1, \quad a_1 \leq m. \quad (1)$$

Distinct octads intersect in at most four points by the Steiner axiom. To identify the possible intersection sizes, fix an octad B and let n_s count the other octads meeting B in s points. Counting j -subsets of B in those octads gives

$$\sum_{s=j}^4 \binom{s}{j} n_s = \binom{8}{j} (\lambda_j - 1) \quad (1 \leq j \leq 4).$$

Solving successively for $j = 4, 3, 2, 1$ yields

$$n_4 = 280, \quad n_3 = 0, \quad n_2 = 448, \quad n_1 = 0, \quad n_0 = 30.$$

Thus every distinct pair of octads intersects in 0, 2, or 4 points.

Lemma 1. *With $f_d = \min\{\binom{d}{2}, 21\}$, one has*

$$3a_2 \leq \sum_{d=3}^{21} f_d a_d. \quad (2)$$

Proof. If $d(T) = 2$, write B, C for the selected octads containing T . Their intersection $Q = B \cap C$ has four points. Exactly two selected octads contain Q , since any further one would contain T .

Each of the other three triples in Q has weight at least three: a triple of weight two there would have the same signature as T .

Different weight-two triples produce different four-subsets Q . Indeed, if two produced the same Q , the two selected octads containing Q would give the same signature to both triples. Charge one unit to each of the three other triples of every such Q .

A receiving triple U of weight $d \geq 3$ lies in only $24 - 3 = 21$ four-subsets of V . It also receives at most $\binom{d}{2}$ charges: each charged four-subset is the intersection of a pair of selected octads containing U , and one pair cannot have two different intersections. Hence U receives at most f_d charges. Counting all charges proves (2). $\square$

Put

$$X = N - a_0 - a_1, \quad E = \sum_{d=3}^{21} (d-2)a_d = 56m - 2N + 2a_0 + a_1,$$

and, for $3 \leq d \leq 21$, set

$$b_d = 6(d-2) - f_d - 3, \quad g_d = \frac{(d-2)(d-3)}{2}.$$

We will use

$$b_d \geq 0, \quad 0 \leq g_d \leq 2b_d. \quad (3)$$

For $d = 3, \dots, 7$, the values of b_d are 0, 3, 5, 6, 6, and those of $2b_d - g_d$ are 0, 5, 7, 6, 2. For $8 \leq d \leq 21$, $b_d = 6d - 36 \geq 12$ and

$$2b_d - g_d = \frac{-d^2 + 29d - 150}{2} \geq 9;$$

the last quadratic is concave and equals 9 at both endpoints. This proves (3) throughout the required range.

Adding $3 \sum_{d=3}^{21} a_d$ to (2) and rearranging gives

$$R := 6E - 3X \geq \sum_{d=3}^{21} b_d a_d \geq 0. \quad (4)$$

By (1),

$$0 \leq R = 336m - 15N + 15a_0 + 9a_1 \leq 345m - 15(N-1). \quad (5)$$

Since $345 \cdot 87 < 15 \cdot 2023$, every separating family has $m \geq 88$.

2.2 Intersecting block pairs and second moments

Let q count unordered pairs of selected octads meeting in four points. Counting triples contained in pairs of selected octads gives

$$\begin{aligned} 4q &= \sum_{d=0}^{21} \binom{d}{2} a_d = X + 2E + \sum_{d=3}^{21} g_d a_d \\ &\leq X + 2E + 2R = 14E - 5X \\ &= 784m - 33N + 33a_0 + 19a_1 \\ &\leq 803m - 33(N-1). \end{aligned} \quad (6)$$

Here the inequality uses (3) and (4).

For each point x , let r_x be the number of selected octads containing x . For each unordered pair $\{x, y\}$, let $z_{xy} = z_{yx}$ be the number containing that pair. If p counts selected octad pairs meeting in two points, expansion of the squares gives

$$\sum_x r_x^2 = 8m + 4p + 8q, \quad \sum_{x < y} z_{xy}^2 = 28m + 2p + 12q.$$

Subtracting half the first equality from the second gives

$$24m + 8q = \sum_{x < y} z_{xy}^2 - \frac{1}{2} \sum_x r_x^2. \quad (7)$$

For each x one has $\sum_{y \neq x} z_{xy} = 7r_x$. Cauchy–Schwarz on the 23 terms, followed by summation over x , therefore yields

$$\sum_{x < y} z_{xy}^2 \geq \frac{49}{46} \sum_x r_x^2.$$

Also $\sum_x r_x = 8m$, so another application of Cauchy–Schwarz in (7) gives

$$24m + 8q \geq \frac{13}{23} \sum_x r_x^2 \geq \frac{13}{23} \frac{(8m)^2}{24} = \frac{104}{69} m^2.$$

Thus

$$q \geq \frac{13}{69} m^2 - 3m. \quad (8)$$

Combining (6) and (8), with $N = 2024$, gives the necessary condition

$$52m^2 - 56235m + 4606371 \leq 0. \quad (9)$$

The left side equals 60379 at $m = 88$ and 13348 at $m = 89$. Together with (5), this excludes all integers below 90 and proves the lower bound of Theorem 1.

3 An exact upper certificate

Appendix A specifies the 127 selected octads as 24-bit masks and gives a complete Python verifier using only integer arithmetic and the standard library. Bit $x - 1$ represents point x . Its design-generation rule is as follows. Over $\mathbb{F}_2$, put

$$g(t) = 1 + t + t^5 + t^6 + t^7 + t^9 + t^{11}.$$

For each polynomial $a(t)$ of degree at most 11, take the 23 coefficients of $a(t)g(t)$ in degrees 0 through 22, and append their parity as coordinate 24. Retain the supports of weight eight. This rule specifies finite data; no coding-theoretic classification is needed to verify that these data form a design.

The verifier generates 4096 distinct words and 759 distinct octads. It checks all $\binom{24}{5} = 42504$ five-subsets, each occurring exactly once, which establishes the $S(5, 8, 24)$ axioms directly. It then checks that the listed 127 masks are distinct members of that design and constructs the signature of each of the 2024 triples by direct containment. All 2024 signatures are distinct. The resulting weight distribution is

d	0	1	2	3	4	5	6	7	8	9
a_d	1	94	441	567	442	283	134	50	10	2

There are no higher weights. The total weight is $7112 = 56 \cdot 127$, and the unique zero-signature triple is $\{3, 16, 18\}$. The distribution is a consistency check; injectivity is checked on the individual signatures. Relabeling preserves containment, so the stipulated uniqueness of the Witt design transfers this construction to $\mathcal{W}$ and proves $m(\mathcal{W}) \leq 127$.

Limits and reproducibility. The lower proof is universal and does not rely on a search over families. The upper certificate checks one feasible family against every triple. The construction was found through greedy selection and local exchange/tabu search; those searches provide no exclusion certificate for smaller families. In particular, this work does not exclude 126 octads, establish optimality of 127, or determine the exact minimum within [90, 127].

Computational and AI assistance. OpenAI Codex assisted with proof development, search and verification code, and manuscript preparation. A separate model context reviewed the lower argument and independently checked the saved design and all triple signatures. This was an additional computational and mathematical review pass, not journal peer review. The decisive finite checks use exact integers; no floating-point result or unverified solver infeasibility claim is used. The complete verifier below reproduces the upper certificate without rerunning the search.

References

- [1] Zijian Zeng, *A Sharp 3-Degree Erdős–Ko–Rado Theorem for 4-Uniform Families*, 2026. <https://doi.org/10.2139/ssrn.7379380>.
- [2] Zijian Zeng, *Improved Lower Bound for the Finite Zarankiewicz Number $z(13, 23; 3, 3)$* , 2026. <https://doi.org/10.2139/ssrn.7386838>.

A The family and a complete verifier

Save the following listing as `verify_certificate.py` and run `python3 verify_certificate.py`, without the `-O` option. Each six-digit hexadecimal token in `HEX` encodes one octad; the order of the tokens is the signature-coordinate order. The listing is also supplied as a separate executable source file. Assertions test the design axioms, membership, and pairwise distinctness of all signatures.

```
"""Exact certificate; run with Python 3, without optimization (-O)."""
from collections import Counter
from itertools import combinations
import json

HEX = """
001f25 0042de 00d847 013532 01cc1a 028c69 02b223 0302ad
03e4a0 4e5108 049616 04d4c8 063b10 064784 085bc0 148545
20a31a 08e036 0a10cb c4b208 48168c 0e60d0 0c0ab4 0cb142
0f0819 10323c 1205a3 0b7802 133091 135320 13aa08 1400f9
149a60 156830 191e10 1a4146 1d8806 2026a6 206478 208275
2934c0 21890d 50c486 a02c45 24d301 260293 a24621 272884
28a521 2b2230 2b4181 611c06 6a6404 32026a 32c508 388838
d12310 3c2064 21514a 406c23 909c0c 451740 45290a 464816
469051 4902d1 4b1421 50228b 506055 95200d 521514 121a86
58a10c 5c0950 5d0228 460ac8 604a85 61224c 088387 64460a
7408a2 6c4031 31d880 708191 725003 805718 806952 8241e8
84e061 850d14 b60406 1c06c2 82b8c0 896411 8ae300 8ca484
```

```

780216 912c82 915016 950162 4c0526 9b0203 9c0c21 a0120f
81019b 230564 a59044 a60870 a80b11 a81434 a1448c b45028
c0a1a2 c13043 c19490 c20ca4 c28918 c38260 092c2c cc10e0
496160 d2420c d88a80 9e8048 e06109 e109c0 819b02
"""

def weight(w):
    return bin(w).count("1")

def points(w):
    return tuple(x for x in range(1, 25) if w >> (x-1) & 1)

g = sum(1 << j for j in (0, 1, 5, 6, 7, 9, 11))
words = []
for a in range(4096):
    w = 0
    for j in range(12):
        if a >> j & 1:
            w ^= g << j
    w |= (weight(w) % 2) << 23
    words.append(w)
assert len(set(words)) == 4096
D = sorted(w for w in words if weight(w) == 8)
assert len(D) == len(set(D)) == 759
assert all(0 <= w < (1 << 24) for w in D)
fives = Counter(t for w in D for t in combinations(points(w), 5))
assert len(fives) == 42504 and set(fives.values()) == {1}
F = [int(h, 16) for h in HEX.split()]
assert len(F) == len(set(F)) == 127 and set(F) <= set(D)
triples = list(combinations(range(1, 25), 3))
signatures = []
for T in triples:
    t = sum(1 << (x-1) for x in T)
    signatures.append(sum(1 << i for i, w in enumerate(F)
                        if t & w == t))
assert len(signatures) == len(set(signatures)) == 2024
hist = Counter(weight(s) for s in signatures)
assert [hist[d] for d in range(10)] == [1,94,441,567,442,283,134,50,10,2]
assert sum(weight(s) for s in signatures) == 7112
zero = [T for T, s in zip(triples, signatures) if s == 0]
assert zero == [(3, 16, 18)]
print(json.dumps({"design_blocks": len(D), "five_subsets": len(fives),
                  "five_subset_multiplicity": 1, "selected_blocks": len(F),
                  "distinct_signatures": len(set(signatures)), "zero_triples": zero,
                  "weight_histogram": dict(sorted(hist.items()))}, indent=2))

```